

LAB RESOURCE UTILIZATION PLATFORM

TEAM 4



Introduction

Laboratories in educational institutions and research organizations contain valuable equipment that is shared among students, researchers, and faculty members. Managing these resources manually through registers, spreadsheets, or paper-based records often leads to scheduling conflicts, inefficient equipment utilization, maintenance delays, and difficulty in tracking resource availability.

The **Lab Resource Utilization Platform** is a web-based application designed to digitalize laboratory resource management. It provides a centralized system where users can view equipment availability, reserve laboratory resources, manage bookings, and monitor equipment utilization efficiently. The platform also enables administrators to maintain equipment records, schedule maintenance, and generate reports for better resource planning.

Project Understanding

The primary purpose of this project is to replace manual laboratory resource management with an automated and centralized platform. The system allows different users to access the application according to their roles and perform tasks such as equipment management, booking, maintenance tracking, and utilization monitoring.

The application will be developed using **React.js** for the frontend, **Spring Boot** for the backend, and **PostgreSQL** as the database. User authentication and authorization will be implemented using **Spring Security** and **JWT** to ensure secure access.

Project Objectives

The objectives of the project are:

- Develop a centralized platform for laboratory resource management.
- Maintain a complete inventory of laboratory equipment.
- Provide real-time equipment availability information.
- Enable online booking and scheduling of laboratory resources.
- Reduce scheduling conflicts and manual work.
- Maintain equipment maintenance and calibration records.
- Improve laboratory resource utilization.
- Implement secure role-based authentication and authorization.
- Generate reports to support effective decision-making.

Project Scope

The project includes the following modules:

- User Registration and Authentication
- Role-Based Access Control
- Equipment Inventory Management
- Equipment Booking and Scheduling

- Equipment Utilization Monitoring
- Maintenance and Calibration Management
- Reports and Analytics

The system is intended for use by educational institutions, research laboratories, and organizations that require efficient management of laboratory resources.

Expected Benefits

The proposed system provides several benefits, including:

- Centralized management of laboratory resources.
- Improved utilization of expensive laboratory equipment.
- Faster and more efficient booking process.
- Reduced scheduling conflicts.
- Better maintenance planning and tracking.
- Secure access based on user roles.
- Improved transparency and operational efficiency.
- Data-driven decision-making through reports and analytics.

Existing System

Many institutions manage laboratory resources using manual registers, spreadsheets, emails, or paper-based booking systems. Equipment availability, booking approvals, and maintenance records are maintained separately, making the overall management process time-consuming and prone to errors.

Problems in the Existing System

The existing laboratory resource management process faces several challenges:

- Manual equipment booking and approval process.
- Scheduling conflicts due to multiple booking requests.
- Lack of real-time equipment availability.
- Poor utilization of laboratory resources.
- Difficulty in tracking maintenance and calibration records.
- Limited reporting and utilization analysis.
- Lack of centralized and secure access management.

Proposed Solution

The proposed **Lab Resource Utilization Platform** provides a centralized web-based solution for managing laboratory resources. The system enables secure user authentication, equipment inventory management,

online booking, utilization monitoring, maintenance tracking, and report generation. It helps institutions improve operational efficiency while reducing manual effort.

Advantages of the Proposed System

- Centralized laboratory resource management.
- Real-time equipment availability.
- Efficient online booking system.
- Reduced scheduling conflicts.
- Improved equipment utilization.
- Better maintenance management.
- Secure role-based access.
- Data-driven reporting and decision-making.

Expected Outcome

The proposed system will replace manual laboratory management with an automated and secure platform. It will improve equipment utilization, simplify booking and maintenance processes, reduce administrative workload, and provide accurate information for efficient laboratory resource planning.

User Roles and Permissions

The Lab Resource Utilization Platform uses Role-Based Access Control (RBAC) to ensure that users can access only the features relevant to their responsibilities. Each user is assigned a specific role with predefined permissions, improving security and efficient management of laboratory resources.

User Roles

Researcher / Student

Researchers and students can search for available laboratory equipment, make booking requests, view booking history, and manage their profile.

Permissions:

- Register and log in
- View available equipment
- Book laboratory equipment
- Cancel own bookings
- View booking history
- Update profile

Lab Technician

Lab Technicians are responsible for maintaining equipment information and monitoring equipment availability.

Permissions:

- View equipment inventory
- Update equipment status
- Record maintenance details
- View booking schedules
- Manage equipment availability

Lab Manager

The Lab Manager supervises laboratory operations and manages equipment bookings.

Permissions:

- Approve or reject booking requests
- Manage equipment inventory
- Monitor equipment utilization
- View maintenance records
- Generate laboratory reports

Department Head

The Department Head monitors laboratory resource utilization within the department.

Permissions:

- View department reports
- Monitor equipment utilization
- View booking statistics
- Review resource usage

Institution Administrator

The Institution Administrator manages laboratory resources across all departments.

Permissions:

- Manage departments
- View institution-wide reports
- Monitor laboratory utilization
- Manage system settings

System Administrator

The System Administrator has complete control over the application.

Permissions:

- Manage users and roles
- Manage equipment records
- Configure system settings
- Access all reports
- Monitor system activity

3. Roles and Permissions Summary

Module	Researcher / Lab Student	Lab Technician	Lab Manager	Department Head	Institution Administrator	System Administrator
User Authentication	✓	✓	✓	✓	✓	✓
Equipment Inventory	View	Update	Manage	View	Manage	Full Access
Equipment Booking	Book	View	Approve/Reject	View	View	Full Access
Maintenance	-	Manage	View	View	View	Full Access
Reports	Own	Basic	Laboratory	Department	Institution	All Reports
User Management	-	-	-	-	Limited	Full Access

Functional Requirements

The system shall provide the following functionalities:

User Authentication

- User registration and login.
- Secure authentication using JWT.
- Role-based access for different users.

Equipment Inventory Management

- Add, update, and remove equipment.
- View equipment details and availability.
- Categorize equipment by laboratory or department.

Booking and Scheduling

- Search available equipment.
- Reserve equipment for a specific date and time.
- Approve or reject booking requests.
- Cancel bookings when required.

Utilization Monitoring

- Track equipment usage.
- View booking history.
- Monitor equipment availability.

Maintenance Management

- Record maintenance details.
- Update equipment maintenance status.
- Maintain service history.

Reports

- Generate equipment utilization reports.
- View booking statistics.
- Generate maintenance reports.

Non-Functional Requirements

Security

- Secure user authentication and authorization using JWT.
- Restrict access based on user roles.

Performance

- The system should respond quickly to user requests.
- Support multiple users simultaneously.

Reliability

- Ensure accurate data storage and retrieval.
- Maintain data consistency during operations.

Scalability

- Support increasing numbers of users and laboratory equipment.

Usability

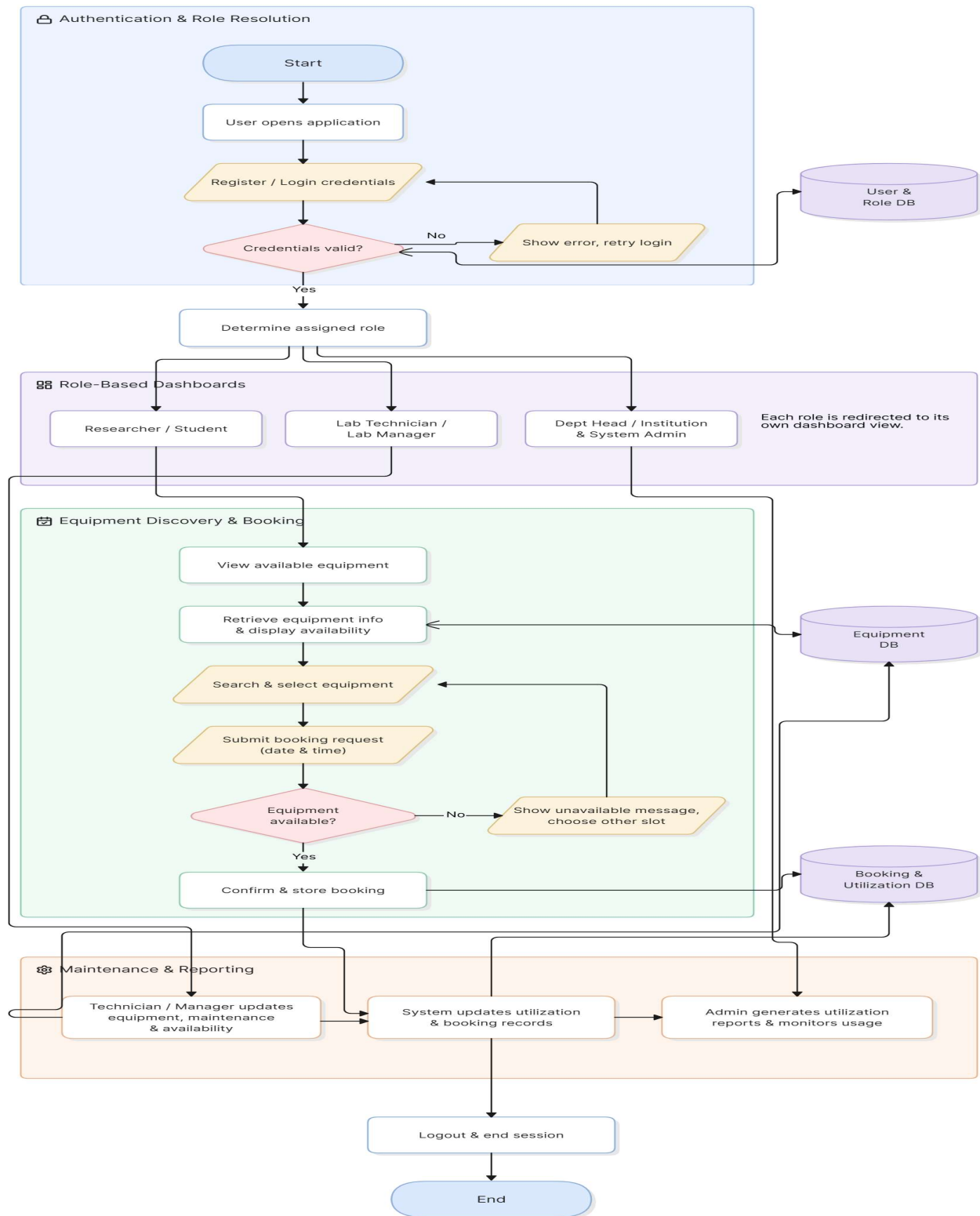
- Provide a simple, responsive, and user-friendly interface.

Availability

- The system should be available whenever users need to access laboratory resources.

Workflow

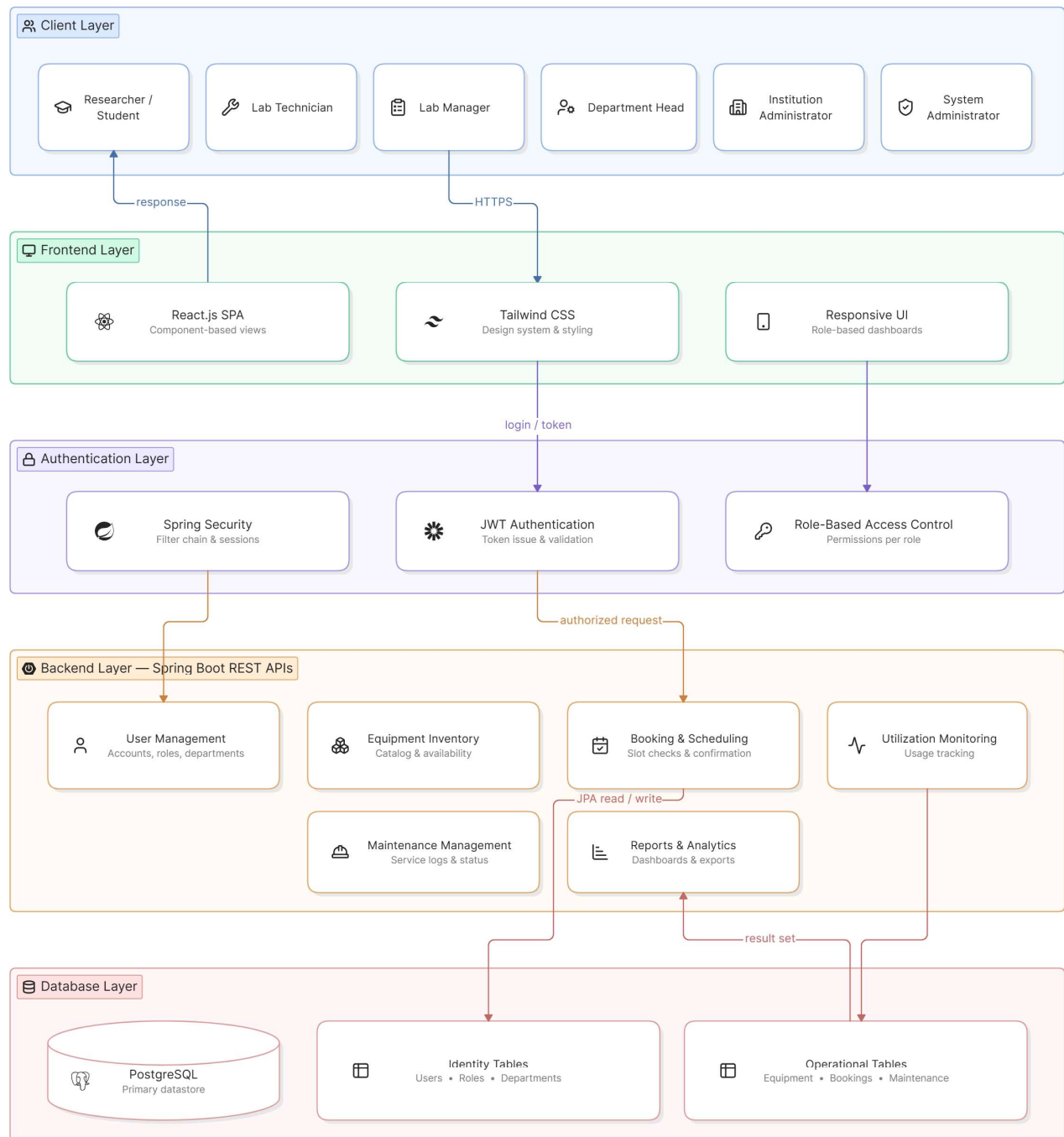
Lab Resource Utilization Platform Workflow / Process Flow



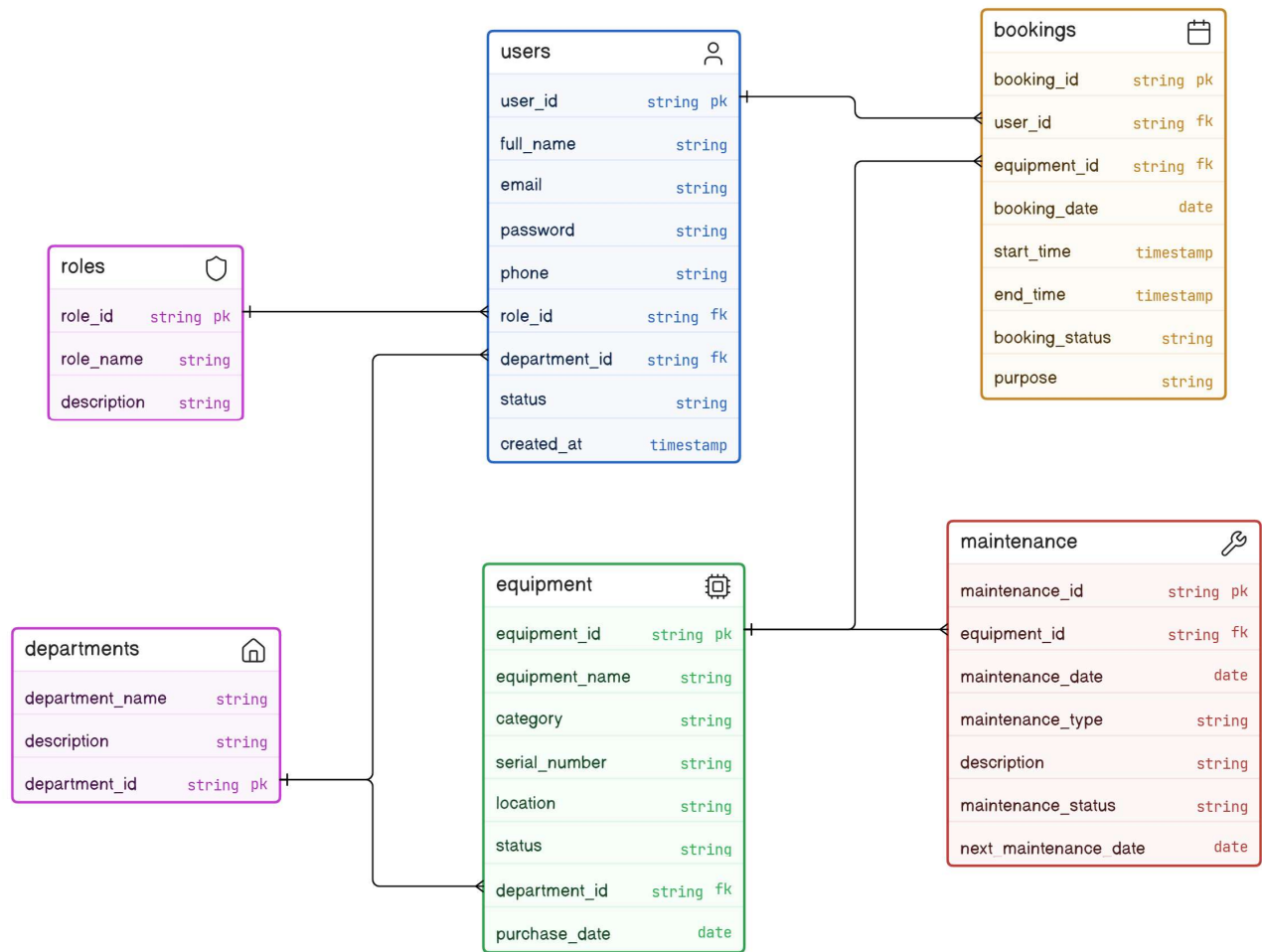
System Architecture

Lab Resource Utilization Platform

High-Level System Architecture



Database Schema & ER Diagram



Overview

The database schema defines the structure of the Lab Resource Utilization Platform and the relationships between different entities. It is designed to provide secure, consistent, and efficient storage of user, equipment, booking, and maintenance information. The database follows a relational model using PostgreSQL to ensure data integrity and minimize redundancy.

Database Tables

Roles

Stores the different user roles in the system, such as Researcher/Student, Lab Technician, Lab Manager, Department Head, Institution Administrator, and System Administrator.

Users

Stores user information including personal details, login credentials, assigned role, and department.

Departments

Maintains details of departments that own or manage laboratory equipment.

Equipment

Maintains information about laboratory equipment, including equipment details, category, department, and current status (Available, Booked, or Under Maintenance).

Bookings

Stores equipment reservation details, including the user, equipment, booking schedule, purpose, and booking status.

Maintenance

Maintains maintenance and calibration records for equipment, including maintenance dates, service details, and current maintenance status.

Entity Relationships

The database consists of the following relationships:

- One Role can be assigned to many Users.
- One Department can have many Users.
- One Department can manage many Equipment items.
- One Equipment Category can contain many Equipment items.
- One User can create many Bookings.
- One Equipment can have many Bookings.
- One Equipment can have many Maintenance records.
-

Summary

The database schema provides a structured foundation for the Lab Resource Utilization Platform by organizing data into related tables. It ensures secure storage, efficient retrieval, and proper relationships between users, equipment, bookings, and maintenance records. The schema is designed to support scalability, maintain data integrity, and facilitate future enhancements.

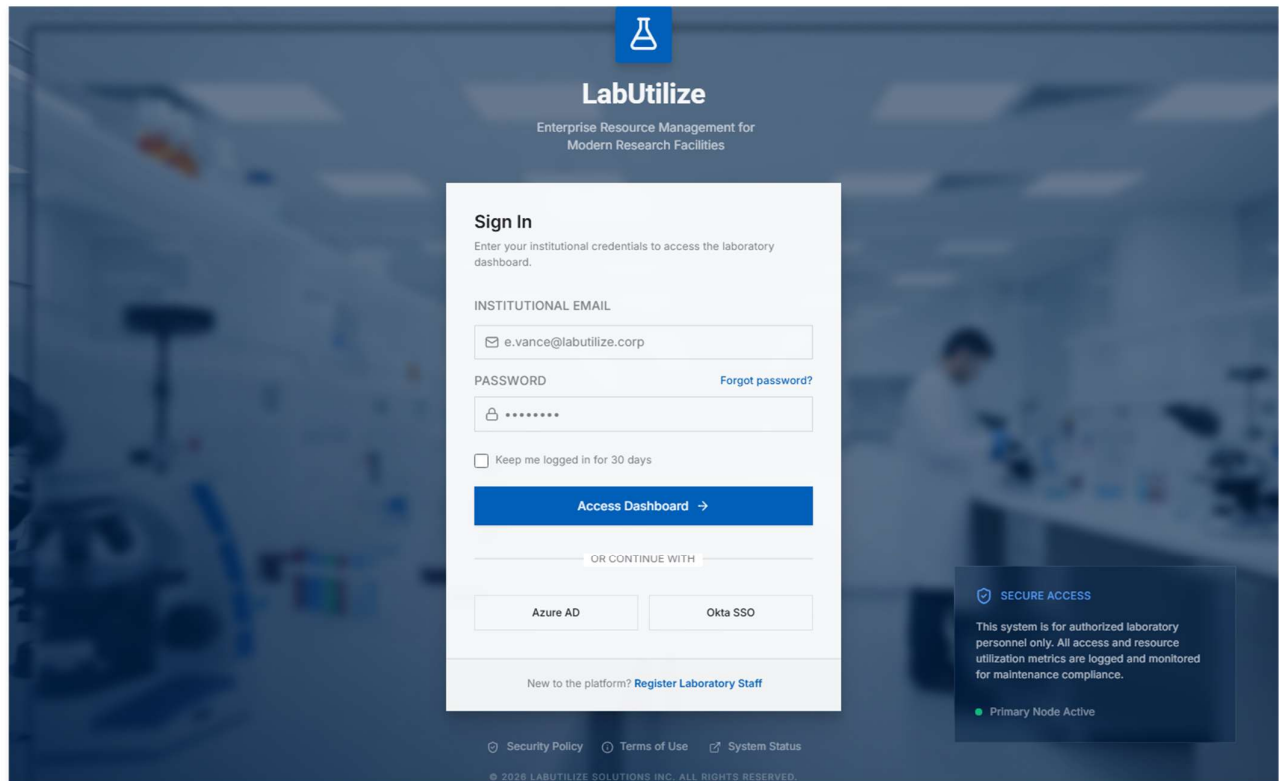
UI Design & Wireframes

Overview

The Lab Resource Utilization Platform is designed with a simple, responsive, and user-friendly interface. The application enables users to efficiently manage laboratory resources through role-based dashboards and intuitive navigation.

Login Page

Allows registered users to securely log into the system.



The image shows a wireframe for the LabUtilize login page. The background is a blurred laboratory scene. At the top center is a blue square icon with a white flask. Below it, the text 'LabUtilize' is displayed in a large, bold, white font, followed by 'Enterprise Resource Management for Modern Research Facilities' in a smaller, lighter font. The main content area is a white rectangular box with a 'Sign In' heading. Below the heading is a subtext: 'Enter your institutional credentials to access the laboratory dashboard.' There are two input fields: 'INSTITUTIONAL EMAIL' with the placeholder 'e.vance@labutilize.corp' and 'PASSWORD' with a masked password '*****'. A 'Forgot password?' link is next to the password field. Below the fields is a checkbox labeled 'Keep me logged in for 30 days'. A large blue button labeled 'Access Dashboard →' is centered. Below this button is a horizontal line with the text 'OR CONTINUE WITH'. Underneath are two buttons: 'Azure AD' and 'Okta SSO'. At the bottom of the white box is a link: 'New to the platform? Register Laboratory Staff'. At the very bottom of the page, there are three links: 'Security Policy', 'Terms of Use', and 'System Status', followed by a copyright notice: '© 2020 LABUTILIZE SOLUTIONS INC. ALL RIGHTS RESERVED.' On the right side, there is a dark blue box with a shield icon and the text 'SECURE ACCESS'. Inside this box, it says: 'This system is for authorized laboratory personnel only. All access and resource utilization metrics are logged and monitored for maintenance compliance.' Below this text is a green dot and the text 'Primary Node Active'.

Registration Page

Allows new users to create an account.



Optimize Your Laboratory Resources.

Join the industry-standard platform for enterprise research facilities. Streamline equipment scheduling, monitor real-time utilization, and maintain operational excellence.

- Enterprise-Grade Security**
Protocol-compliant data protection for sensitive research.
- Precision Scheduling**
Conflict-free booking for high-demand lab equipment.
- Maintenance Tracking**
Automated service logs and operational health monitoring.

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Create Professional Account

Register to access the LabStream Pro management dashboard.

FULL NAME

INSTITUTIONAL EMAIL

PHONE NUMBER

DEPARTMENT

PASSWORD

Minimum 8 characters with a mix of symbols.

☐ I agree to the [Terms of Service](#) and [Research Privacy Protocol](#).

Register Laboratory Profile →

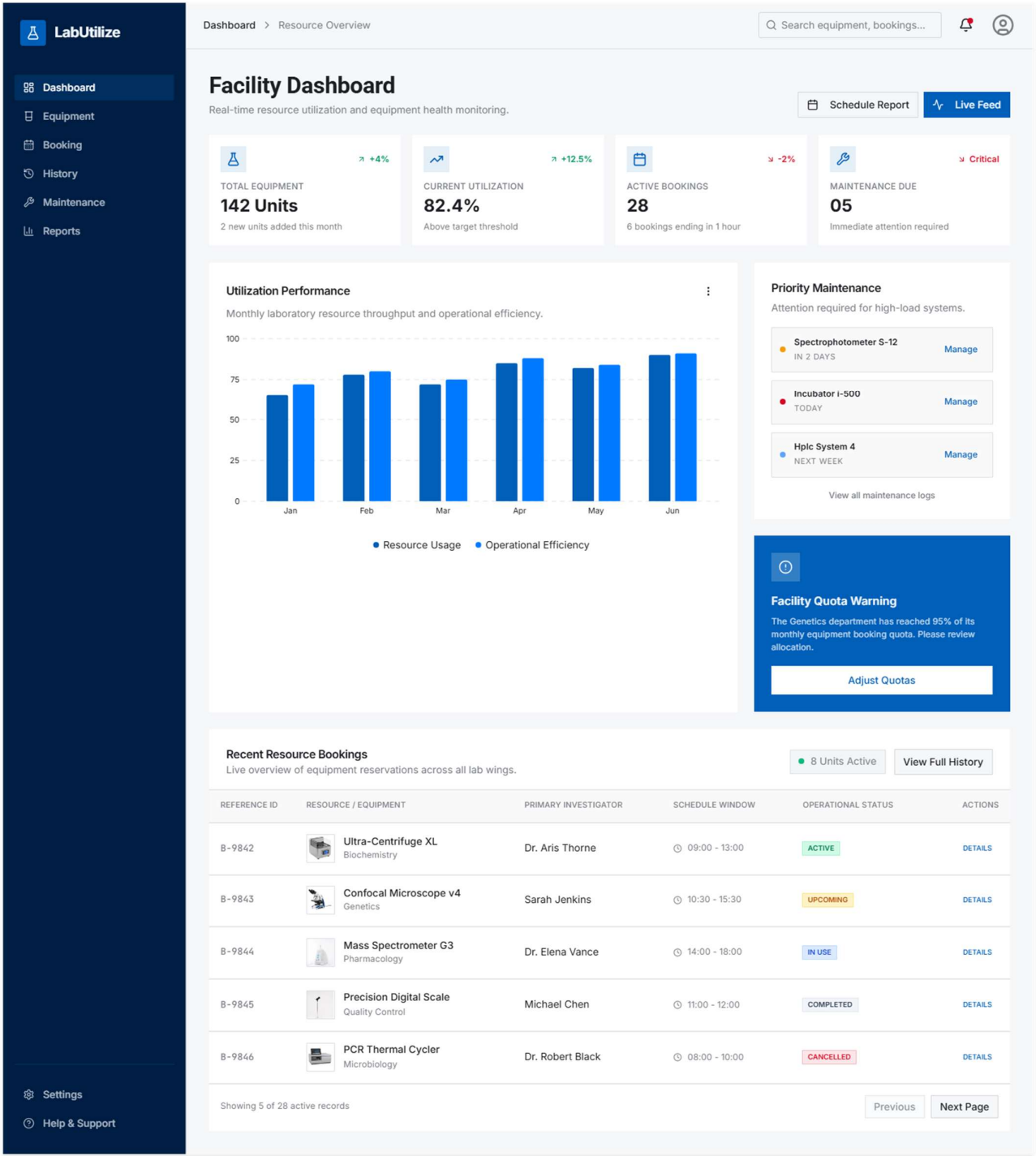
OR ALREADY REGISTERED?

Sign In to Existing Account

For institutional SSO or bulk registration inquiries, please contact your [Facility Administrator](#).

Dashboard

Provides an overview of laboratory resources based on the user's role.



Equipment Management Page

Displays all laboratory equipment and their current status.

LabUtilize

Dashboard

Equipment

Booking

History

Maintenance

Reports

Settings

Help & Support

Dashboard > Resource Overview

Search equipment, bookings...

Equipment Inventory

Manage and track high-value laboratory assets, maintenance schedules, and real-time utilization.

Export CSV

Add New Asset

Total Assets142

Currently In Use38

Pending Service5

Offline / Warning2

Filter by name or S/N...

All Categories

Showing 5 of 5 units

Columns

Image	Asset Name ¹	Serial Number	Category	Location	Status	Utilization	Actions
	High-Speed Refrigerated Centrifuge ID: EQ-4001	CENT-2024-001	Centrifuge	Lab Room 302	AVAILABLE	68%	...
	Advanced Confocal Microscope ID: EQ-4002	MICR-8821-XP	Microscopy	Imaging Suite A	IN USE	92%	...
	Precision Analytical Digital Scale ID: EQ-4003	SCAL-5502-B	Analytical	Chemistry Lab 1	AVAILABLE	45%	...
	Mass Spectrometer System ID: EQ-4004	MS-9910-PR0	Analytical	Core Facility	MAINTENANCE	12%	...
	Ultra-Low Temperature Freezer ID: EQ-4005	FRZ-7700-ULT	Storage	Bio-Storage Unit	AVAILABLE	100%	...

Upcoming Maintenance

Scheduled preventative service for the next 30 days.

View Calendar

Mass Spectrometer System

Technician: John Doe

Mar 24, 2024

CALIBRATION

Advanced Confocal Microscope

Technician: Sarah Smith

Mar 28, 2024

DEEP CLEAN

High-Speed Centrifuge

Technician: Mike Ross

Apr 02, 2024

ROTOR INSPECTION

Resource Insights

Automated utilization summary

Facility Utilization72%

"Peak demand observed between 10:00 AM and 3:00 PM. High congestion in Microscopy suite."

TOP PERFORMERS

Centrifuge B-0194%

Freezer ULT-0488%

Generate Report

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Booking Page

Allows users to reserve laboratory equipment.

LabUtilize

Dashboard

Equipment

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Help & Support

Dashboard > Resource Overview

Search equipment, bookings...

SCHEDULING SYSTEMNode ID: LAB-RESOURCE-RESERVATION-MGR

Equipment Reservation

Select specialized laboratory instrumentation and allocate temporal resources for your research objectives.

Reservation Parameters

Configure the utilization period and technical justification for the resource.

STEP 1 OF 2

RESOURCE IDENTIFICATION

High-Speed Centrifuge XL-500

OPERATIONAL DATESTART SEQUENCEEND SEQUENCE

Time

Time

UTILIZATION PROTOCOL / PURPOSE

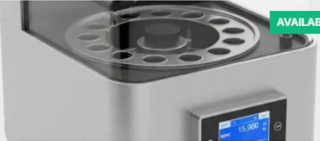
Detail the experimental procedure or study objective...

Estimated Duration: N/A

Cancel

Commit Reservation →

AVAILABLE



High-Speed Centrifuge XL-500

Biolab A - Room 402

TECHNICAL SPECIFICATIONS

Operational Limit25,000 RPM / 50,000 x g

Calibration Date24-OCT-2023

Temp Range-20°C to +40°C

Maintenance CycleQuarterly (Q4 Pending)

UTILIZATION INDEX82% (Peak)

CONFLICTS & AVAILABILITY

MORNING SESSION08:00 - 11:30 | Clear

LUNCH MAINTENANCE12:00 - 13:00 | System Refresh

AFTERNOON SESSION13:30 - 17:00 | Dr. Smith

VIEW FULL SCHEDULE MAP >

Note: Cancellations must be logged 24 hours prior to the reservation start time to avoid resource idle-time penalties. All users must maintain current safety certification.

LabUtilize > Resource Scheduler > Equipment Booking

SYSTEM STATUS: NOMINALCOMPLIANCE: ISO 17025LAST AUDIT: 12-FEB-24

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Booking History Page

Displays all bookings made by the user.

LabUtilize

Dashboard

Equipment

Booking

History

Maintenance

Reports

Settings

Help & Support

Dashboard > Resource Overview

Search equipment, bookings...

Booking History

Comprehensive log of your laboratory equipment reservations. Track usage patterns, download reports, or manage upcoming time slots.

Export CSV

New Reservation

TOTAL BOOKINGS142

ACTIVE SESSIONS3

PENDING APPROVAL5

COMPLIANCE RATE98.4%

All ReservationsUpcomingCompletedCancelled

Search by ID or equipment...

BOOKING ID	EQUIPMENT DETAILS	USER / DEPARTMENT	DATE & TIME	STATUS	ACTIONS
BK-8821	High-Speed Centrifuge X1 LAB-CEN-001	Dr. Elena Vance Biochemistry	Oct 24, 2023 09:00 AM - 11:30 AM	CONFIRMED	
BK-8819	Digital Precision Scale LAB-SCA-012	Marcus Thorne Pharmacology	Oct 23, 2023 02:00 PM - 03:00 PM	COMPLETED	
BK-8815	Advanced Fluorescence Microscope LAB-MIC-045	Dr. Elena Vance Biochemistry	Oct 22, 2023 10:00 AM - 01:00 PM	IN-USE	
BK-8802	High-Speed Centrifuge X1 LAB-CEN-001	Sarah Jenkins Genetics	Oct 21, 2023 08:30 AM - 10:30 AM	CANCELLED	
BK-8798	Digital Precision Scale LAB-SCA-012	Dr. Elena Vance Biochemistry	Oct 20, 2023 04:00 PM - 05:00 PM	COMPLETED	

Showing 1-5 of 142 records

<

1

2

3

...

29

>

Need to reschedule?

Avoid cancellation penalties by rescheduling at least 24 hours in advance. Contact the facility manager for urgent changes.

Contact Manager

View Policy

Download Utilization Report

Generate a comprehensive report of your equipment usage across all projects for current quarter auditing.

Quarterly Audit (Q3-2023).pdf

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Maintenance Page

Allows Lab Technicians to manage equipment maintenance records.

LabUtilize

Dashboard

Equipment

Booking

History

Maintenance

Reports

Settings

Help & Support

Dashboard > Resource Overview

Search equipment, bookings...

Maintenance Management

Monitor asset health, track scheduled service intervals, and review historical maintenance logs.

Filter Assets

New Service Request

SYSTEM HEALTH

92.4% ▲ 1.2%

AT RISK ASSETS

03 Requires Inspection

CRITICAL DOWN

01 ▲ 1

COMPLIANCE SCORE

98.5% SLA Met

Asset Health Monitor

Real-time status of critical laboratory inventory

View All Assets >

Equipment	Current Health	Operational Status	Next Service	Actions
High-Speed Centrifuge X1 EQ-402 • Centrifuge	94%	OPERATIONAL	2024-04-15	...
Precision Spectrometer Alpha EQ-109 • Microscope	62%	DEGRADED	2024-02-28	...
Digital Analytical Balance EQ-882 • Scale	100%	OPERATIONAL	2024-06-01	...
PCR Thermal Cycler v4 EQ-331 • Analyzer	15%	DOWN	2023-11-12	...

SCHEDULED TASKS

SERVICE HISTORY LOGS

High-Speed Centrifuge X1

Calibration

Tomorrow, 09:00

Medium Priority

ASSIGNED TO

Marcus Thorne

PENDING

View Details

Precision Spectrometer Alpha

Repair

Today, 14:00

High Priority

ASSIGNED TO

Marcus Thorne

IN PROGRESS

View Details

PCR Thermal Cycler v4

Repair

Overdue (2 days)

Critical Priority

ASSIGNED TO

Sarah Jenkins

OVERDUE

View Details

Facility Compliance

Regulatory & Safety Standards

Calibration Standard ISO 17025

100%

Scheduled PM Compliance

96.4%

Emergency Repair Latency

82.1%

Certified Compliant

Next audit: March 15, 2024

System Alerts

Critical Hardware Failure

PCR Cycler (EQ-331) reported motor seizure. System shutdown initiated.

12 HRS AGO

Predictive Warning

Vibration levels on Centrifuge X1 are trending above normal. Early service recommended.

2 HOURS AGO

Upcoming Calibration

Calibration due for Analytical Balance. Technician M. Thorne notified.

4 HOURS AGO

View Notification History

Predictive Insights

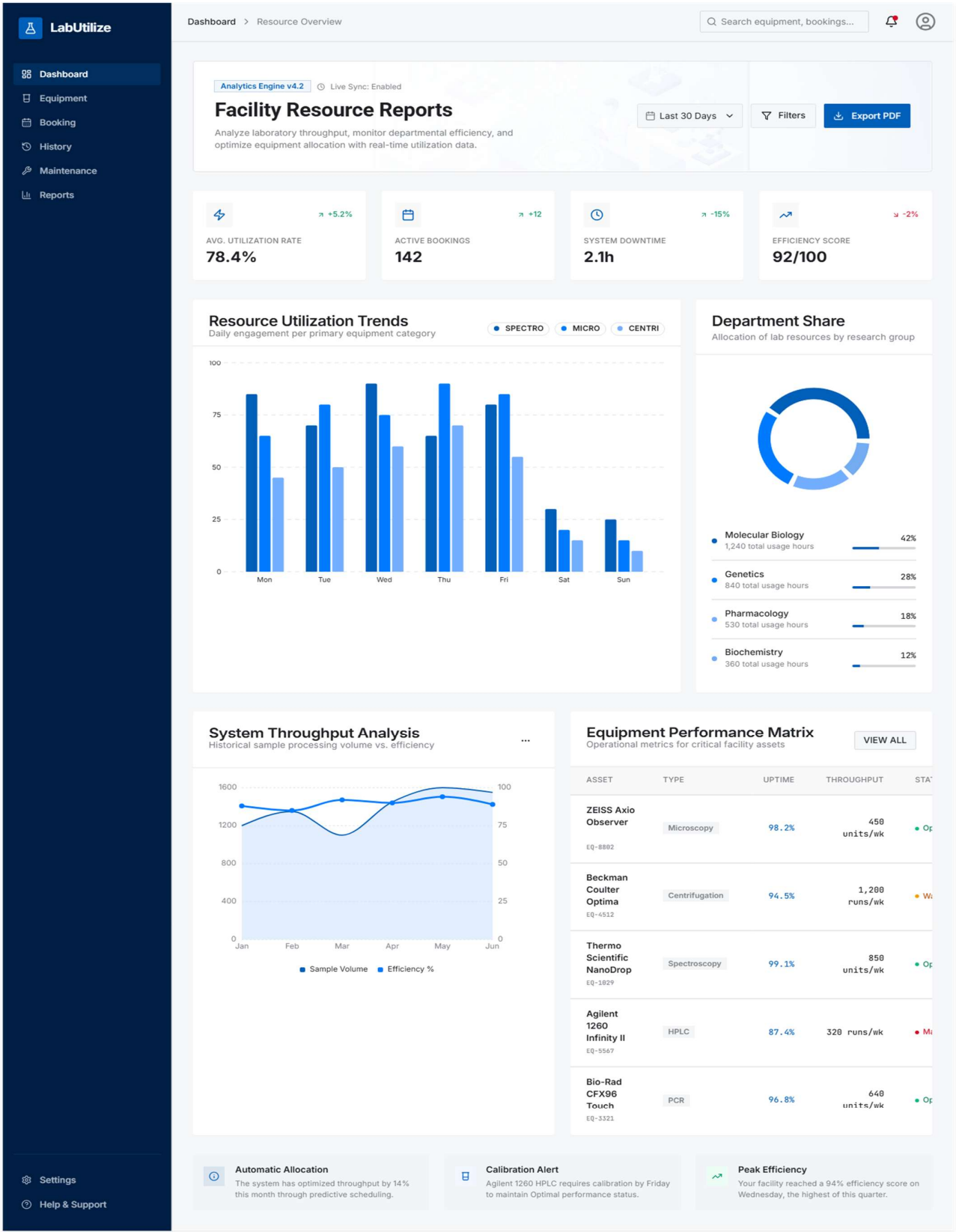
Based on current usage patterns, your lab could save 12% in downtime by rescheduling calibration for 3 high-load assets next week.

Optimize Schedule

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Reports Dashboard

Provides reports for administrators to monitor laboratory resource utilization.



Made with Visily

Wireframe Summary

The user interface is designed to provide easy navigation, secure access, and efficient management of laboratory resources. Each screen focuses on a specific function, ensuring a simple and consistent user experience across the application.